MODULE 1: PHOTONICS

Introduction:

LASER: - Light Amplification by stimulated Emission of Radiation.

Laser was invented by **T. H. Maiman** in the year 1960.

At the beginning it was known as **MASER** (Molar Amplification by stimulated Emission of Radiation).Basically, laser is a light beam, which is coherent in nature, monochromatic (one colour) and highly directional (low divergence).

Properties of LASER light:

1. **Coherence:** Laser light is highly coherent, meaning the waves are in phase with each other, both spatially and temporally. This coherence allows for the creation of very intense and focused beams.
2. **Unidirectionality:** Laser beams travel in a single, well-defined direction with minimal divergence, unlike ordinary light sources that spread out in all directions.
3. **Monochromaticity:** Laser light consists of a single wavelength, or color, unlike ordinary light which is a mixture of many wavelengths. This single wavelength allows for precise control over the interaction of light with materials.
4. **High Intensity:**Laser light is highly intense, meaning it has a large amount of power per unit area, due to the coherence and monochromaticity..

Basic Principle:

Radiation interacts with matter under appropriate conditions. The interaction leads due to transition of the quantum system such as an atom or a molecule from one energy state to another energy state.

If the transition is from a higher state to a lower one, the system gives out a part of energy and if the transition is in the reverse direction, then it absorb the incident energy.

Consider two energy states E_1 and E_2 of a system. If the energy difference between the two energy levels is ΔE.

Then, $\Delta E = E2 - E_1$

Max Planck suggested for electromagnetic radiation of frequency 'ν' as

$$\nu = \frac{\Delta E}{h} = \frac{E_2 - E_1}{h} \qquad \text{------(1)}$$

Where, ν - frequency of radiation,

h - Planck's constant (6.63×10^{-34} Js)

If it is incident on the system of energy state E_1 then by absorbing the energy system move to energy state E_2. On the other hand, If a system is in state of E_2 then it emits an electromagnetic radiation (a photon) of frequency "ν" given by equation (1) then the energy of the system changes to E_1.

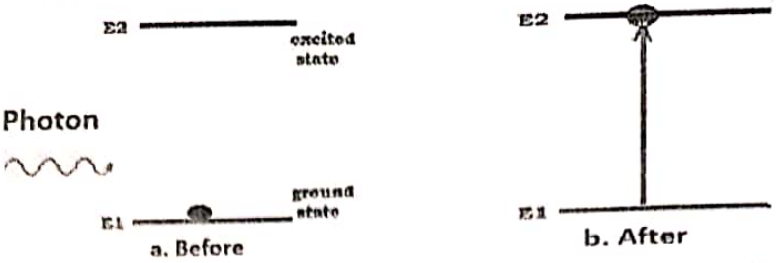
There are three possible ways through which interaction of radiation and matter can take place,

1. Induced Absorption
2. Spontaneous Emission
3. Stimulated Emission

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1. **Induced Absorption:**

Induced absorption is the absorption of an incident photon (energy) by which the atom excited from lower energy state to a higher state.

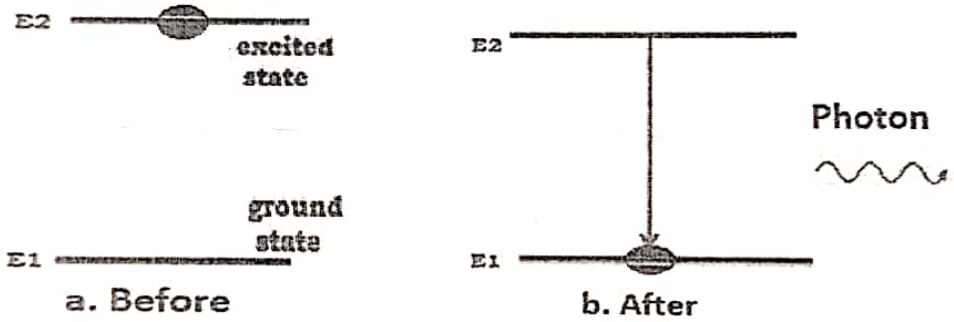


Let E_1 and E_2 be two energy levels in the energy level scheme of an atom in which E_1 corresponds to lower energy and a photon having and energy $\Delta E = E_2 - E_1$ is incident on the atom under which condition the atom absorbs the photon, As a result its energy becomes $E_1 + \Delta E = E_2$ Hence it is shown in the level E_2. Then, the atom is said to have made transition to the excited state. This is called induced absorption (Stimulated absorption, and Absorption)

And represented as, Photon + Atom ⟶ Atom*

2. **Spontaneous Emission:**

Spontaneous emission is the emission of a photon, when a system transits from a higher energy state to a lower energy state to a lower energy state without the aid of any external energy.



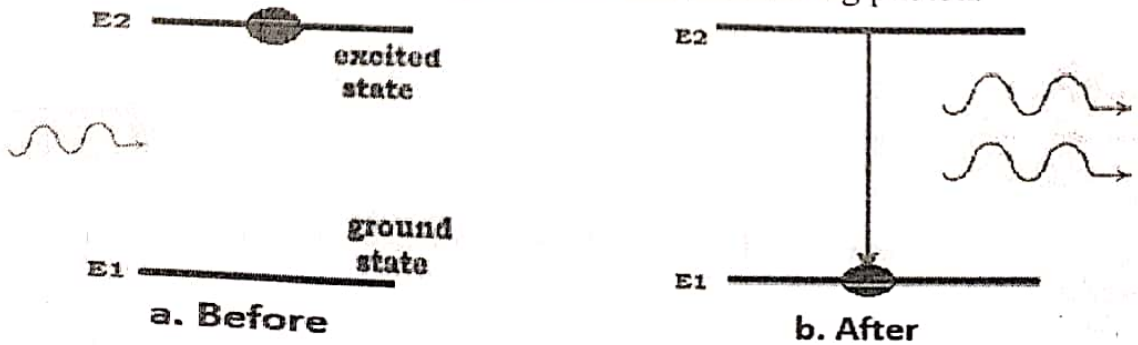
Consider an atom in the excited state. Before of the universal tendency for a system to attain the least available energy state the atom voluntarily emits a photon of energy $\Delta E = E_2 - E_1$. Then the energy of the atom becomes. $E_2 - \Delta E = E_1$. Since the atom emits the photon without being aided by any external means. It is called spontaneous emission.

And represented by, Atom* ⟶ Atom + Photon

3. **Stimulated Emission:**

Stimulated emission is the emission of a photon by a system under the influence of external energy (resonant frequency), due to which the system transits from higher energy state to a lower energy state.

The photon thus emitted is called the stimulated photon and will have same phase, energy and direction of movement as that of the passing photon called stimulating photon.



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Consider an atom in the excited state. Let a photon having an energy $\Delta E = E_2 - E_1$ interact with the atom by passing in its vicinity. Under such stimulation the atom emits a photon and transit to the lower energy state. The two photons travel in exactly the same direction and with exactly the same energy.

The electromagnetic waves associated with the two photons will have identical phase and thus they are coherent and represented by

$$\text{Atom}^* + \text{Photon} \longrightarrow \text{Atom} + (\text{Photon} + \text{Photon})$$

Which is responsible for laser action.

Einstein's A &B Co-efficients

Consider two energy states E_1 &E_2 (where $E_2 > E_1$) of a system of atoms and N_1 &N_2 are number of density of atoms per unit volume of each energy state respectively. Now the energy of photons absorbed and emitted by atoms is given by

$$E = h\nu$$

$$E2 - E_1 = h\nu$$

$$\nu = \frac{E_2 - E_1}{h}$$

Let us consider absorption and two emission processes

1. **Case of Induced absorption:**

   In this, rate of induced absorptions per unit volume, per unit time is called rate of induced absorption and depends on

   Number of density of lower energy state $\mathbf{N_1}$

   and Energy density $\mathbf{U_\nu}$

   $\therefore$ Rate of induced absorption $\alpha\ \mathbf{N_1\ U_\nu}$

   $\therefore$ Rate of induced absorption $= \mathbf{B_{12}\ N_1\ U_\nu} - - - - - - - - (1)$

   Where, $\mathbf{B_{12}}$ is proportionality constant of Einstein's Coefficient of induced absorption.

2. **Case of spontaneous emission:**

   In this, rate of spontaneous emissions per unit volume, per unit time is called rate of spontaneous emission and depends on

   Number of density of higher energy state $\mathbf{N_2}$

   $\therefore$ Rate of spontaneous emission $\alpha\ \mathbf{N_2}$

   $\therefore$ Rate of spontaneous emission $= \mathbf{A_{21}\ N_2} - - - - - - - - (2)$

   Where, A_{21} is proportionality constant of Einstein's Coefficient of spontaneous emission.

3. **Case of stimulated emission:**

   In this, rate of stimulated emission per unit volume, per unit time is called rate of stimulated emission and depends on

   Number of density of higher energy state $\mathbf{N_2}$

   and Energy density $\mathbf{U_\nu}$

   $\therefore$ Rate of stimulated emission $\alpha\ \mathbf{N_2\ U_\nu}$

   $\therefore$ Rate of stimulated emission $= \mathbf{B_{21}\ N_2\ U_\nu} - - - - - - - - (3)$

   Where, $\mathbf{B_{12}}$ is proportionality constant of Einstein's Coefficient of stimulated emission.

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Under thermal equilibrium the total energy remains unchanged, hence

Rate of induced absorption = Rate of spontaneous emission + Rate of stimulated emission

∴ from equations 1, 2, and 3

$$B_{12} N_1 U_\nu = A_{21} N_2 + B_{21} N_2 U_\nu$$

$$U_\nu (B_{12} N_1 - B_{21} N_2) = A_{21} N_2$$

$$U_\nu = \frac{A_{21} N_2}{(B_{12} N_1 - B_{21} N_2)}$$

Dividing by B_{21} and re-arranging we get

$$U_\nu = \frac{A_{21}}{B_{21}} \left[\frac{1}{\frac{B_{12} N_1}{B_{21} N_2} - 1} \right] \text{--------} (4)$$

From Boltzmann's law we have

$$\frac{N_2}{N_1} = e^{\frac{-h\nu}{kT}}$$

$$\therefore \quad \frac{N_1}{N_2} = e^{\frac{h\nu}{kT}} \text{--------} (5)$$

Substituting in equation 4 we get

$$U_\nu = \frac{A_{21}}{B_{21}} \left[\frac{1}{\frac{B_{12}}{B_{21}} e^{\frac{h\nu}{kT}} - 1} \right] \text{--------} (6)$$

According to Planck's radiation law

$$U_\nu = \frac{8\pi h\nu^3}{c^3} \left[\frac{1}{e^{\frac{h\nu}{kT}} - 1} \right] \text{--------} (7)$$

Comparing equations 6 & 7 we get

$$\frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3} \quad \text{and} \quad \frac{B_{12}}{B_{21}} = 1 \quad \Rightarrow \quad B_{12} = B_{21}$$

Which implies probability of induced absorption is equal to probability of stimu

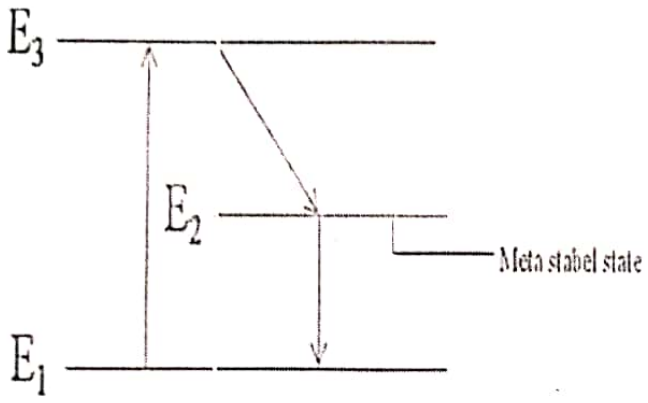
emission. Hence A_{21} & B_{21} can be simply as A & B

Hence at thermal equilibrium the equation for density is

$$U_\nu = \frac{A}{B} \left[\frac{1}{e^{\frac{h\nu}{kT}} - 1} \right]$$

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- **Population inversion:**



It is a state in which more number of atom will be present in higher energy state compared to anyone of a lower energy state.

OR

If by some means number of atom in the excited state is made to exceed number of atom in the ground state then population inversion is established between the excited state and ground state

Metastable state:

Consider E1 is the ground state of atom. E2 and E3 are excited state or normal state in which E2 is the metastable state. The population of energy levels are N1, N2,N3 respectively.

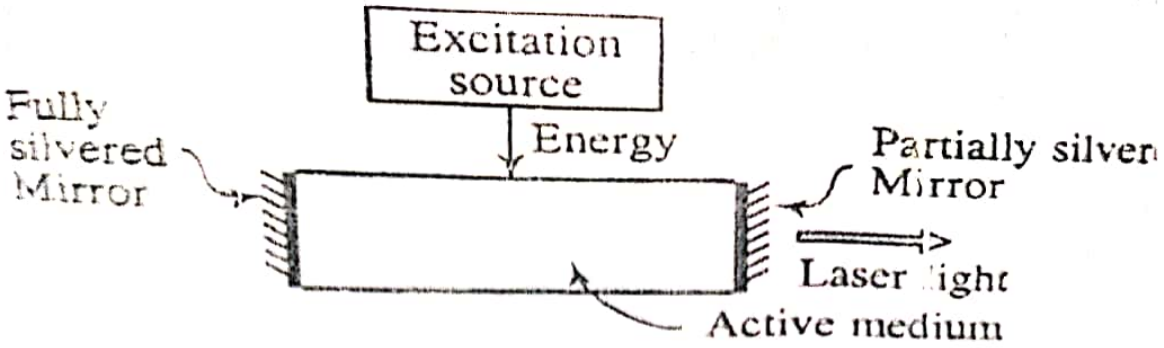
A photon with energy hγ=E3-E1 is incident, thereby atom directly move from E1 to E3 level. As E is the excited state, within a short interval of time (10^{-2} s)atom jumps back to lower energy level. A E2 is metastable state, the atom reaching to E2 to stay for the period(10^{-3} to 10^{-2})s. This process wi be continuous and thereby we have population inversion between E2 and E1.

Requisites of a laser system:

The three requisites of laser system are,

1. An excitation source for pumping action
2. Active medium

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1.Pumping unit or excitation unit:

Appropriate amount of energy is to be supplied for the atom in order to excite them to higher energy state. The energy input may be in the form of a light energy then it is called optical pumping. If it is in the form of electrical energy then pumping is called electrical pumping.

2.Active medium:

The active medium in a laser system may be in a solid, gas, semiconductor. Population inversion occur at certain stage in active medium due to absorption of energy. After this stage the active medium is capable of emitting a laser light.

3.Laser cavity or resonance cavity:

Usually laser cavity is constructed by placing two reflecting mirror across the active medium. The laser cavity provides the feedback necessary to tap certain permissible part of laser energy from active medium.

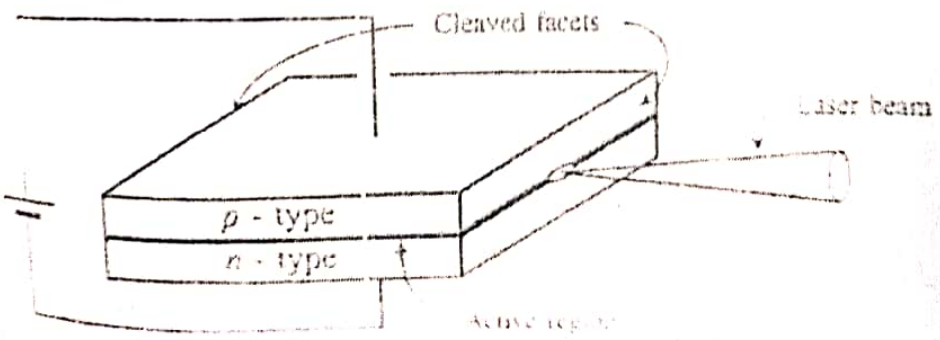
- ## Semiconductor laser:

Semiconductor laser is the one in which the active medium is formulated by semiconducting materials.

Construction:

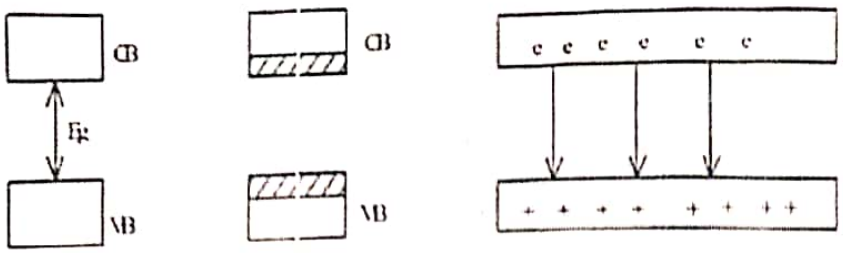
It consists of a single crystal of gallium arsenide and consist of heavily doped n & p section. The n section is formed by doping a tellurium where p section doping by zinc. The doping concentration is very high an order of 10^{17} to 10^{19} dopant atom /cm^3.

An external battery is connected in such a way that PN junction form is always maintained in forward bias. Finally to have a laser cavity two surface of crystal is polished such that one surface act as a full reflecting &other partially reflecting surface.



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- Working:-



a b c

When a semiconductor material such as Gallium Arsenide placed at absolute 0^0 K, behaves like perfect insulator. On increasing the temperature of that semiconductor and maintaining forward voltage across junction a small current will flow across PN junction.

On further increases forward voltage current across the junction also increases indicating that electrons are now moving from valence band into conduction band within a short interval of a it time electrons enter into conduction band are move to lower energy level of conduction band, when current across junction reaches to threshold value, population inversion is achieved between lower energy level of conduction band with respect to top energy level of valence band as shown in figure (b).Now, if an electron's jumps from conduction band into valence band or in other words if an electron recombines with holes produce proton. This photon initiates stimulated emission process. Thus photon produced by stimulated process are move between reflecting surface till they build up laser light. Finally the laser light comes out of a partially reflecting surface.

The wavelength laser light given by,

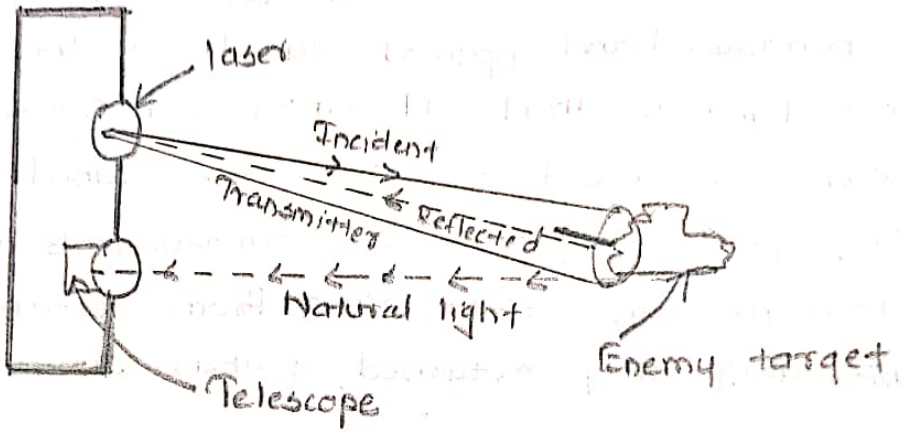
$$\lambda = hc/Eg$$

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APPLICATION OF LASER :-

By the virtue of their high intensity, high degree of monochromaticity & coherence, lasers find remarkable applications such as medicine, communications, material processing, energy resources, 3-D photography etc.

1) Laser rangefinder in defense :-



Laser Range finder In Action

The principle of laser range-finder is same as that of conventional radar. The excellent directionality of laser light makes an ideal device for ranging purpose. In defense, it is important to have accurate data about the distance of enemy target to have the 'first hit' chance.

In today's military, all battle tanks are fitted with laser rangefinder interfaced with computers to provide information in a digital readout form. The rangefinders are also used for continuous tracking & ranging of missiles & aircraft from ground, or form air.

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Working:-

A high powered pulsed laser from solid state laser device such as Nd-YAG laser is directed towards the enemy target from transmitter. The pulses are narrow with high peak power. Upon incidence, the beam bounces from surface of target as a reflection. (from fig.) Inside the receiver, there will be an interference filter. It is narrow-band optical tuned to the frequency of laser light. so that all background noise entering the receiver is wiped off. Then the signal is amplified by using photomultiplier. The rangefinder's high speed clock measures the exact time from instant the pulses left unit until they returned, & then converts it into distance.

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Use of Attenuators for Single-Photon Sources

In quantum optics and quantum communication, it is often necessary to use light at the level of single photons. However, standard lasers emit many photons per pulse, which is unsuitable for such experiments. Optical attenuators are devices designed to reduce the intensity of light in a controlled manner.

Attenuators can be of two types:

- **Fixed attenuators** – which reduce light intensity by a constant factor.
- **Variable attenuators** – where the transmission can be adjusted.

Role of single photon attenuators in Quantum Communication:

- To reduce the optical power of laser pulses.
- To ensure only one photon is present per pulse.
- To control multi-photon pulses
- The proper attenuation ensure the mean photon number

By carefully reducing the laser output, attenuators allow the probability of multi- photon pulses to become very low, thereby approximating a single-photon source. Such sources are essential for quantum key distribution (QKD), quantum cryptography, and photon correlation experiments.

[Definition: A single photon source is a device that emits only one photon at a time on demand a probalistic.

Optical attenuators are the devices that reduce the power level of optical signal without changing its characteristics such as wavelength or waveform.]

Optical Modulators

Optical modulators are devices that control the properties of light such as amplitude, phase or polarization, usually by applying an external electric field. Two important mechanisms are based on electro-optic effects are Pockels effect and Kerr effect.

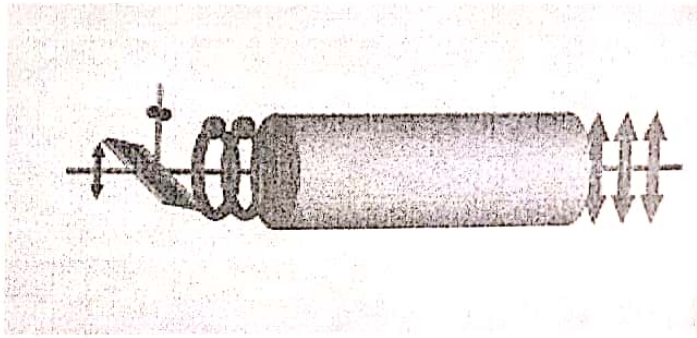
Pockels Effect (Linear Electro-Optic Effect)

Certain crystals, such as Lithium Niobate and Potassium Dihydrogen Phos- phate exhibit a change in refractive index that is linearly proportional to the applied electric field.

When polarized light passes through such a crystal under an applied voltage, its phase changes. With the help of polarizers, this phase modulation can be converted into amplitude modulation, allowing the light to be switched on or off.

Applications:

- High-speed switching in optical communication.
- Optical shutters in laser experiments



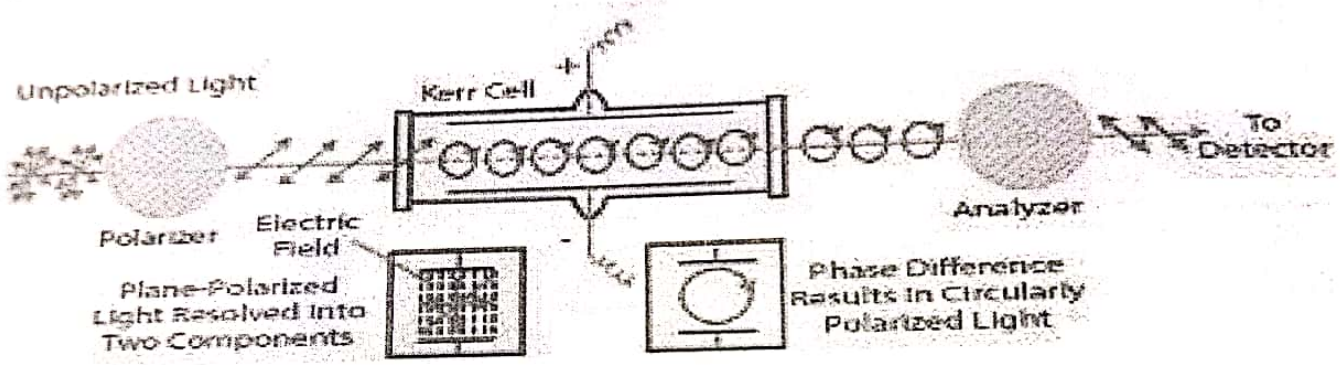
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Kerr Effect (Quadratic Electro-Optic Effect)

In certain materials, the refractive index changes quadratically with the applied electric field. This phenomenon, called the Kerr effect, is weaker than the Pockels effect but can occur in symmetric materials where the Pockels effect is absent. A Kerr cell can rotate the polarization of light depending on the voltage applied.

Applications:

- Ultrafast optical switching in laser systems.
- Phase and amplitude modulation in optical communication.



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OPTICAL FIBERS

Historically the first experimental proof that light could be guided was provided by the British physicist John Tyndall. He demonstrated to the royal society that light can be guided along a curved stream of water employing the sell known optical phenomenon of total internal reflection. The first glass fiber was made in the 1920s but the concept of cladding brought forth in the 1950 made practical the guiding of light with limited success.

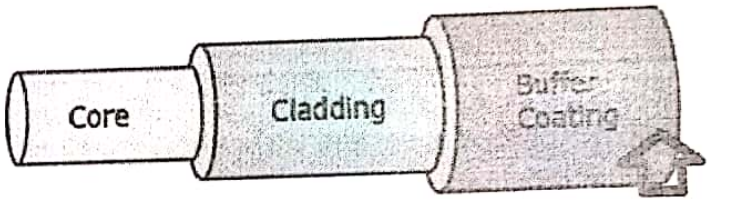
The fiber optics communication is also known as light wave communication, but the name PHOTONICS is becoming increasingly popular Charles. K. Kao. father of fiber optics defines photonics as follows.

"PHOTONICS is the enabling technology for optical fiber communication to improve its performance and to reduce its cost"

Optical fibers are essentially light guides used in optical communications as waveguides. They are transparent dielectrics and able to guide visible and infrared light over long distances.

Construction:

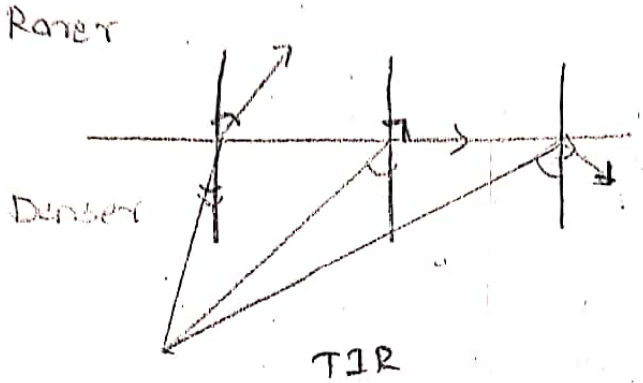
An optical fiber is cylindrical in shape and it has just two parts. The inner part which is a cylindrical material is made of glass or plastic and is called the core. The outer part of the core surrounded by a hollow cylinder called cladding. The cladding is also made of similar material but of lesser refractive index. The cladding is enclosed in a polyurethane jacket, which safeguards the fiber against chemical reaction with surroundings, and also against crushing. Many such fibers, each one protected by individual jacket, are grouped to form a cable. A cable may consist of one to several hundred such fibers.



Parts of an Optical fiber

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Principle :- It is based on principle of Total internal reflection. Consider a ray of light passing from denser medium to rarer medium. As angle of incidence ↑ the angle of refract^n. also ↑. for a particular angle of incidence called critical angle the refracted ray just grazes the interface (Angle of refraction is 90°) If angle of incidence is greater than critical angle than ray reflected back to denser medium. This phenomenon is called TIR.



TIR

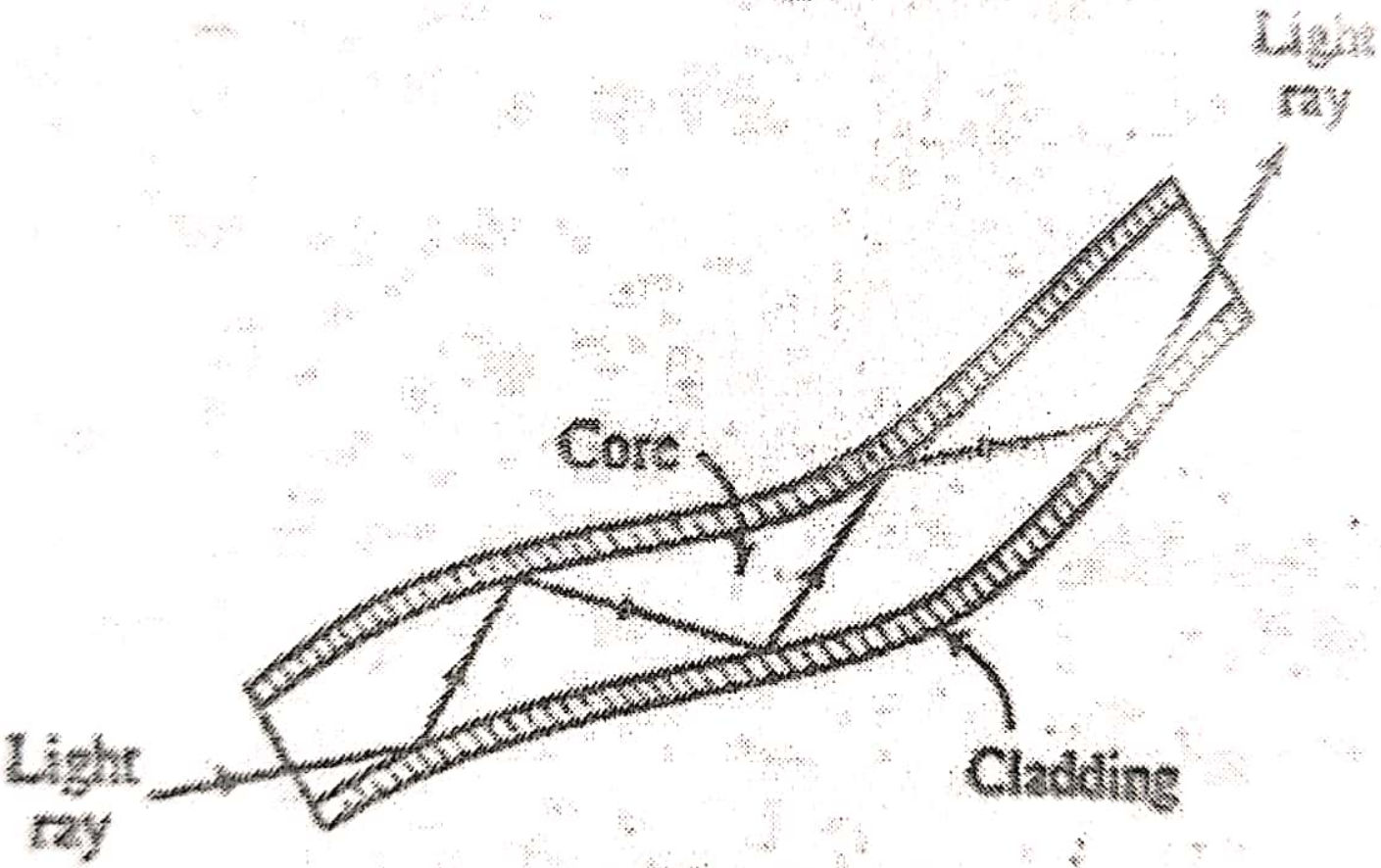
* Propagation Mechanism (OR working principle) :-

The main f^n. of the optical fibre is to accept maxi. light & transmit the same with mini. attunation (reduct^n. of force effect),

The incident light enter the core & strikes the interface of the core & cladding at large angle as shown fig. In optical fibre, the refractive index of cladding is always lesser than that for its core. The light suffers total internal reflection. This is possible since by geometry the angle of incidence at the interface is greater than critical angle. The signal sustain (comfort, help, strengthen) its strength & also confines (keep) itself completely within core during propagation. Thus optical fibre f^n. as a waveguide.

The propagation continues even if the fiber is bent ~~is~~ but not too sharply. (Since, for sharp bends, the light fails to undergo TIR & signal strength drops drastically!

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FIBER AS A WAVEGUIDE

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Acceptance angle & Numerical Aperture:

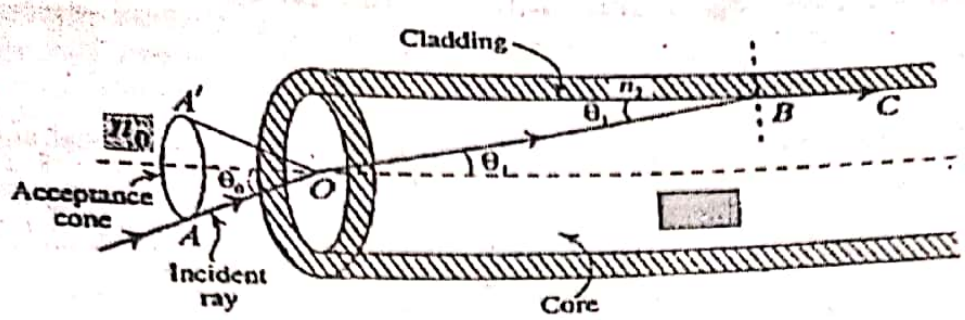
The maximum external incident angle for which the light will propagate in fiber is called acceptance angle. OR It is the angle of incidence at the axis of optical fiber such that light moves along the interface of core and cladding.

The angle θ_0 is called the waveguide acceptance angle or the acceptance cone half-angle, and $\sin\theta_0$ is called the numerical aperture (N.A) of the fiber. The numerical aperture represents the light-gathering capability of the optical fiber.

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Condition for Propagation:

Angle of Acceptance & Numerical Aperture :



ACCEPTANCE CONE

Let $n_0, n_1, and\ n_2$ be the refractive indices of surrounding medium, core of the fiber, and cladding respectively.

Now, for refraction at the point of entry of the ray into the core,

By applying Snell's law we get

$$n_0 \sin\theta_0 = n_1 \sin\theta_1, ---------(1)$$

At the point of interface

The angle of incidence = $90 - \theta_1$

Again applying Snell's law, we have,

$$n_1 \sin(90 - \theta_1) = n_2 \sin 90.$$

$$n_1 \cos\theta_1 = n_2$$

$$\cos\theta_1 = \frac{n_2}{n_1} \quad ------(2)$$

Rewriting Equation (1) we have,

$$\sin\theta_0 = \frac{n_1}{n_0}\sin\theta_1$$

$$\sin\theta_0 = \frac{n_1}{n_0}\sqrt{(1 - cos^2\theta_1)}$$

Substituting for $\cos\theta_1$ from equation (2) we have

$$\sin\theta_0 = \frac{n_1}{n_0}\sqrt{1 - \frac{n_2^2}{n_1^2}} = \frac{n_1}{n_0}\sqrt{\frac{n_1^2 - n_2^2}{n_1^2}}$$

$$\boldsymbol{\sin\theta_0 = \frac{\sqrt{n_1^2 - n_2^2}}{n_0}}$$

If the surrounding medium of the fiber is air then n_0=1

$$\sin\theta_0 = \sqrt{n_1^2 - n_2^2}$$

i.e., $\mathbf{N.A = \sqrt{n_1^2 - n_2^2}}$

If θ_i is the angle of incidence of an incident ray, then the ray will be able to propagate,

if $\theta_i < \theta_0$

or, if $\sin\theta_i < \sin\theta_0$

or, $\sin\theta_i < \sqrt{n_1^2 - n_2^2}$

This is the condition for propagation. ***i.e.,*** $\boldsymbol{\sin\theta_i < N.A}$

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V-Number:

The number of modes supported for propagation in the fiber is determined by a parameter called V-number (denoted as V). V-number is given by,

$$V = \frac{\pi d}{\lambda}\frac{\sqrt{n_1^2 - n_2^2}}{n_0} \quad or, \quad V = \frac{\pi d}{\lambda}(NA), \quad \text{For air } V = \frac{\pi d}{\lambda}\sqrt{n_1^2 - n_2^2}$$

Where, d is the core diameter, n_1 is the refractive index of the core, n_2 is the refractive index of the cladding, n_0 is the refractive index of surrounded medium and λ is the wavelength of light propagating in the fiber.

For V>>1, the number of modes supported by the fiber (approximately) is given by,

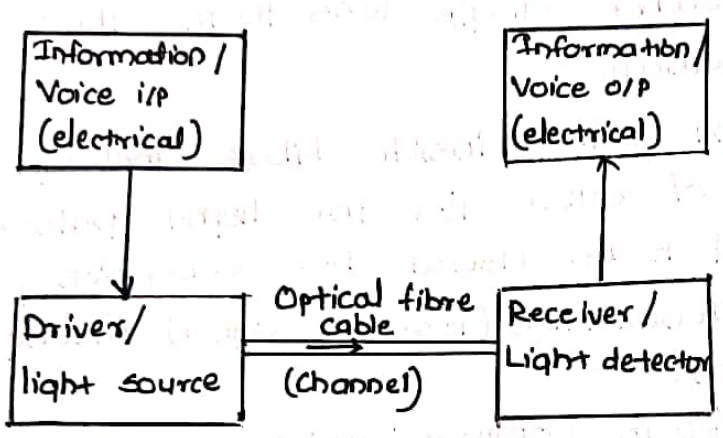
$$number\ of\ modes \cong \frac{V^2}{2}$$

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* Application Of Optical Fibers:-

Application of optical fibres find ... technology was introduced ... ds. Each strip ... communication, data transfer, audio/video transfer, medical, engineering, entertainment etc. Here, discussed point to point communications in detail.

Point - to - point communication:-



Point-to-point communication

A simple point to point communication system using optical fibres as shown in fig.

It consist of three basic components -

1) An optical transmitter i.e the driver/light source to transmit the signal/pulses.
2) The communication medium (channel) i.e optical fibre, &
3) An optical receiver, usually a photo detector, to convert light pulses back into electrical signal.

The information/voice to be transmitted will be in an analog electrical signal format. An optical fibre can receive & transmit signal only in form of optical pulses.

The function of driver/light source is to work as an transducer to convert i/p electrical signal into suitable light pulses. As LED or laser used as

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consists of

The electrical signal is fed to semiconductor laser system, & get modulated to generate an equivalent digital sequence of pulses which turn laser on & off. This form a series of optical pulses representing i/p information which is coupled into optical fibre cable of an incidence angle less than that of acceptance cone angle.

Next the light pulses inside fibre undergo TIR & reach other end of cable. For the light pulses to reach the end, & to be useful for reception & conversion, the attenuation (loss in signal strength) should be minimum.

The final step in communication system is to receive the optical signals at the end of the optical fibre & convert them into equivalent electrical signals.

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Attenuation :-

Attenuation is the loss of power suffered by optical signal as it propagates thr! the fiber. It is also called the fiber loss.

The three mechanisms thr! which attenuation take place are 1) Absorption 2) Scattering 3) Radiation losses.

1) Absorption :-

In this case, the loss of signal power occurs due to absorption of photons associated with signal. There are photon absorbed by two types.

a) **Absorption by impurities :-**

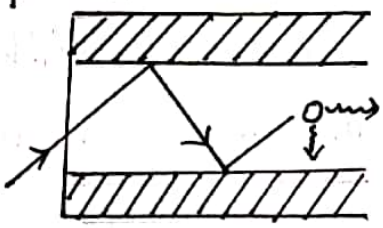
This type of impurities that are present in fiber glass such as iron, chromium, cobalt & copper. During signal propagation, photons will be absorbed by impurities & this by get excited to higher energy level during de-exicition process photon are emitted by with usually different wavelength or at least in different phase. Hence it is a loss.

b) **Intrinsic :-** As the core medium construced with high pure or plastic, so when light signal is propagated into optical fiber some of photons will be get absorb by the core medium. This is called as intrinsic absorption.

2) **Scattering losses :-** Scattering loss will be observed in optical fiber only when their is sharp change in RI. When light signal is propagated into optical fiber & if it reaches to the region, where there is change in RI exist from which photon will undergo scattering as shown fig.

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This type of scattering of photons leads to loss in the optical fiber.

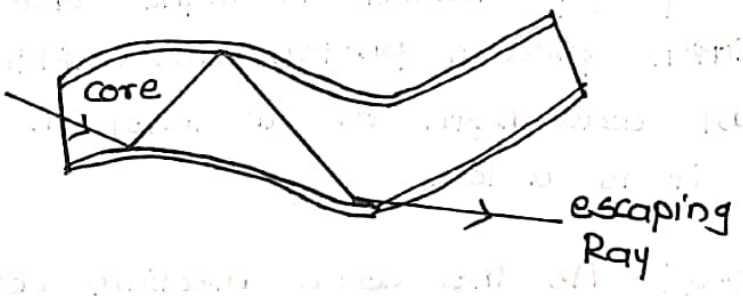


The scattering of the photons due to change in R.I of an optical fiber is similar to relay scattering which has characteristics property of λ^4.

3) Radiation Loss:-

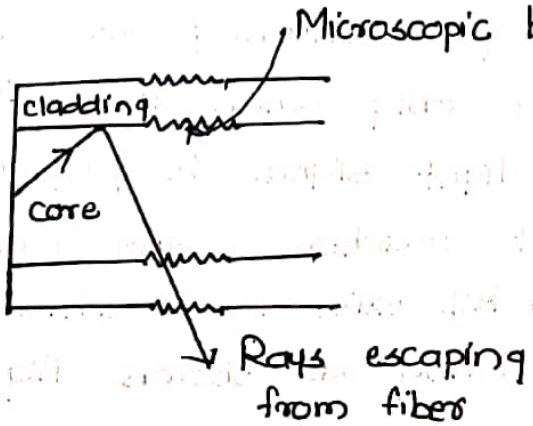
There are two types of radiation loss & they are

a) Macrobends :- These bends occur while wrapping the fiber on a spool or turning it around corner. The loss will be negligible for small bends. If optical fiber is bend more than some critical value, then there will be formation of macrobends as shown fig. When light signal reaches to macrobend region, from there instead of TIR, it may undergoes refraction & thus leads to loss.



b) Microscopic Bends :-

Due to application of unequal pressure on the surface of optical fiber, forms microscopic bends as shown fig. from that part signal undergoes refraction instead of TIR.



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